

## EEE 306 (January 2025)

Power System I Laboratory

### Final Project Report

Section: C2 Group: 03

## Fault Classification and Location Detection of VSC-HVDC Transmission System

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### Course Instructors:

Azazul Islam, Lecturer  
Mostaqul Islam, PT

Signature of Instructor: \_\_\_\_\_

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|                                                                              |                                                                        |
|------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Signature: _____<br>Full Name: Ratul Das<br>Student ID: 2106176              | Signature: _____<br>Full Name: Tonmoy Deb<br>Student ID: 2106177       |
| Signature: _____<br>Full Name: Anannya Chowdhury Moon<br>Student ID: 2106178 | Signature: _____<br>Full Name: Saumik Banik<br>Student ID: 2106179     |
| Signature: _____<br>Full Name: Asibur Rahman Rakib<br>Student ID: 2106180    | Signature: _____<br>Full Name: Mushfiqur Rahman<br>Student ID: 2106181 |

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# 1 Abstract

**Voltage Source Converter-based High Voltage Direct Current (VSC HVDC) systems** address the modern world's need for sustainable, efficient, and reliable energy by enabling efficient long-distance transmission, flexible grid connections, and enhanced stability. However, fault detection in HVDC transmission systems is crucial to prevent system failure, as faults can cause significant energy loss and potentially collapse the transmission system. VSC-HVDC technology, expected to be heavily used in future power systems, is more prone to faults than Alternating Current (AC) transmission, necessitating protection by accurately detecting fault type and location. This project investigated different types of AC faults and one DC fault at five exact locations along a VSC-HVDC transmission line using an equivalent Simulink model to simulate real-life faults. The generated data were used to train various machine learning (ML) and deep learning (DL) models for fault classification. While single models showed limited accuracy (highest **73.74%** with **Random Forest**), a **three-stage CNN** achieved the best results classifying normal vs. AC vs. DC faults with **93.15%** accuracy, AC faults at **86.57%**, and DC faults at **80.18%**. For fault location detection, regression methods applied on a separate test dataset yielded the best results with **Gaussian Process Regression**. In both classification and DC fault location detection, accuracy was lower for DC faults, indicating that further work on DC fault detection is necessary.

## 2 Introduction

The rapid transformation of modern society is marked by increasing energy demands, widespread urbanization, and the integration of renewable energy sources. To meet these challenges, energy transmission systems must evolve to become more sustainable, efficient, and reliable. Voltage Source Converter-based High Voltage Direct Current (VSC HVDC) transmission systems play a vital role in enabling long-distance power transmission, flexible grid interconnections, and stable integration of renewables. However, their adoption introduces a complex engineering problem that involves advanced technology, sophisticated control, and robust protection strategies.

The complexity of VSC HVDC systems arises from their dependence on advanced power electronics and digital control, which must operate reliably under varying and often harsh electrical conditions. The system must quickly detect, classify, and locate faults (both AC and DC) to prevent equipment damage, widespread power outages, or grid instability. The nonlinear behavior of power electronic converters, real-time signal processing requirements, and the need for high-speed, accurate fault response add further layers of difficulty. This makes the protection and control of VSC HVDC systems an inherently complex, multidisciplinary engineering challenge.

The importance of addressing these challenges is heightened by the critical nature of the transmitted energy. Any failures in rapid detection and mitigation of faults can lead to

catastrophic consequences, including large-scale blackouts and substantial losses in energy and infrastructure.

There are different possible ways to solve these problems. One of them is traditional protection scheme, which involves usage of hardware-based relays. Fault current behavior in HVDC systems is different from AC, with lower impedance and the absence of natural zero crossings, making traditional distance protection less effective. [1](#) Different types of advanced control strategies are continuously being adopted for effective fault condition control. [2](#) Different signal processing techniques such as wavelet transformation method are used for fault detection, classification. [3](#)

However, these methods often suffer from limitations in speed, accuracy, and the ability to adapt to evolving system conditions. Moreover, the increasing complexity and scale of modern HVDC networks demand more sophisticated and responsive fault management strategies. In recent years, the integration of AI in electrical power systems has opened new horizons for addressing complex challenges that were previously intractable with conventional methods. AI, particularly machine learning and deep learning, offers unprecedented capabilities in pattern recognition, anomaly detection, and predictive analytics, making it an ideal candidate for enhancing fault detection and mitigation strategies in HVDC systems.

This project aims to investigate the success of different AI models in detecting faults, classifying those faults, then also detecting the location of faults.

## 3 Design

### 3.1 Problem Formulation (PO(b))

#### 3.1.1 Identification of Scope

The scope of this project focuses on solving the key issues related to **fault detection and location identification** in a VSC-based HVDC transmission system. Specifically, this project aims to:

- **Accurately classify different fault types** (AC faults: LL, LG, LLG, LLL; DC faults: LL, LG) and distinguish them from normal operating conditions.
- **Determine the precise location of DC faults** along the HVDC transmission line, which is essential for quick isolation and system protection.
- **Develop a reliable dataset** from a detailed VSC-HVDC Simulink model that captures fault variations in terms of type, resistance, and time windows.
- **Provide a framework for future enhancements**, such as hybrid CNN-LSTM models, improved feature engineering, and dataset enrichment, to achieve higher accuracy and real-time applicability.

#### 3.1.2 Literature Review

- **Data Extraction Source:**

The data for fault diagnosis in VSC-HVDC systems was primarily extracted from simulation platforms such as PSCAD/EMTDC. Lan et al. [4] collected voltage signals from both rectifier and inverter stations. Ding et al. [5] worked with post-fault current signals which were further processed using Fast Fourier Transform (FFT) and Variational Mode Decomposition (VMD). Another study [6] employed the Stockwell transform to extract time-frequency features from simulated internal and external AC and DC fault signals. Meta-heuristic ANN-based approaches [7] gathered fault data characterized by variations in resistances, fault locations, and inception angles, optimized for feature selection.

- **Dataset:**

The datasets varied in size and fault diversity. Lan et al. [4] created a comprehensive dataset of 32,000 voltage signal samples sampled at 100 kHz, incorporating various fault types, locations, and resistances from 50 to 3500  $\Omega$ . Ding et al. [5] used about 3,360 samples representing 10 fault classes across different fault distances, split between training on 11 distances and testing on 10 unseen distances. The Bi-GRU-based study [6] utilized 3,628 internal DC faults, 8,870 external DC faults, and 1,960 AC fault cases. Meta-heuristic ANN

approaches [7] worked with multiple fault classes featuring optimized sets of fault-related parameters, though exact dataset sizes were not detailed.

- **Classification Models:**

Deep learning models were widely applied for fault classification. Lan et al. [4] designed a two-stage 1D Convolutional Neural Network (1D-CNN) that performed fault classification effectively. Ding et al. [5] also employed a 1D-CNN, which outperformed traditional classifiers like Support Vector Machines (SVM) and Random Forest. The 2024 research [6] applied a bi-directional Gated Recurrent Unit (Bi-GRU) network that showed resilience against noise and parameter changes. Additionally, meta-heuristic optimized Artificial Neural Networks (ANNs) used algorithms such as Grey Wolf Optimizer (GWO), Particle Swarm Optimization (PSO), and Cuckoo Search Optimization (CSO) to enhance classification, achieving up to 100% accuracy [7].

- **Regression Models:**

For fault location estimation, Lan et al. [4] utilized the same two-stage 1D-CNN model, reporting an average location error of 0.168 km and reliable identification of high-resistance faults up to 5.2 k $\Omega$ . Meta-heuristic ANN approaches [7] combined ANN-based classification with Gaussian Process Regression (GPR) for fault location, demonstrating improved regression accuracy over conventional linear regression models. The other studies [5], [6] focused mainly on classification and did not implement regression models for location estimation.

### **Model for Classification and Regression Problem:**

#### **Classification:**

- **CNN:** The crucial advantage of CNNs is that both feature extraction and classification operations are fused into a single machine learning body to be jointly optimized to maximize the classification performances. [8](#)
- **ANN:** The ANN is a good online and offline method to achieve fast fault detection, determining fault types, and finding fault location in overhead transmission lines. [9](#)
- **ANN using GWO, PSO, CSO algorithms:** The results of this three-algorithm combination recorded full score accuracy, which can enhance fault detection and classification. [10](#)
- **Random Forest:** It can handle non-linear relationships and interactions between features, can provide insights into the importance of different features in fault detection and location. [11](#)

#### **Regression:**

- We are doing regression using Interactive Linear Regression, Stepwise Linear Regression, Neural network, and Gaussian Process Regression. As all of them showed promising result in relevant researches. [12](#)

### 3.1.3 Formulation of Problem

Our project focuses on detecting and locating faults in a VSC HVDC system, addressing several limitations found in previous research. These earlier works faced the following challenges:

- **No Stepwise Fault Separation:** They did not apply a systematic approach to differentiate between AC and DC faults, which could improve detection accuracy.
- **Limited Fault Location Methods:** Fault location was either ignored or approached with only one method, missing the opportunity to explore multiple models for better results.
- **Restricted Simulation Tools:** They relied on less flexible tools and did not utilize MATLAB Simulink for detailed system modeling and analysis.
- **Simplistic Datasets:** The datasets lacked structured time-window tracking of fault evolution, reducing their effectiveness in capturing fault dynamics.

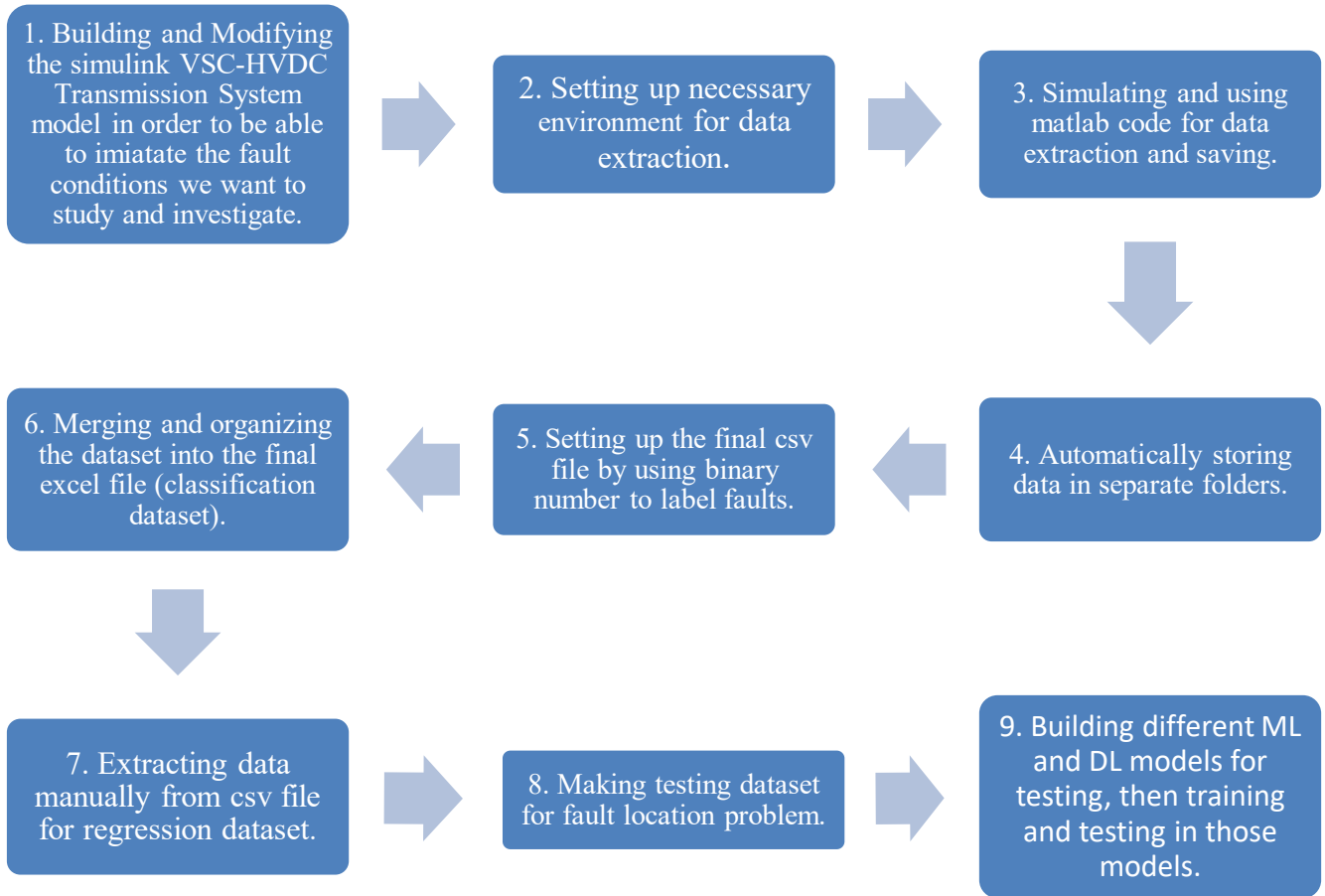
### 3.1.4 Analysis

To address the formulated problem, several key aspects were analyzed to ensure the successful design and implementation of the fault classification and location detection system:

1. **Built a Model in Simulink:** We created a VSC HVDC system in MATLAB Simulink with AC System I, AC System II, Station 1 (AC to DC), Station 2 (DC to AC), a Pi line (with resistance, inductance, capacitance), three-phase faults, line-to-line faults, and a controller with PLL and power control. This let us test faults at both ends.
2. **Made a Dataset with One-Hot Encoding:** We built a dataset with 2400 samples and 27 columns, using 5 fault locations, 10 resistance levels, and 48 time windows. It has 24 features, plus fault resistance, time window, and distance (in km). We used one-hot encoding to turn fault types into numbers, making it easy for our models to learn.
3. **Tried Basic ML and DL Structures:** We used simple machine learning (like Random Forest) and deep learning (like ANN and CNN) as our base. This gave us a strong start to build our fault detection and location models.
4. **Designed a 3-Stage CNN Classifier:** We made a 3-stage CNN to spot faults. It first checks if there's a fault, then sorts AC or DC faults, and finally digs into DC fault details. Our tests showed DC faults were tricky (54% for ANN, 74% for CNN), and this method helped improve detection.
5. **Tested Four Models for Fault Location:** We checked four models—Linear Regression, Stepwise Regression, Neural Network, and Gaussian Process Regression (GPR)—to find fault spots. GPR did the best, but accuracy varied across distances.
6. **Used a Test Dataset for Predictions:** We tested a separate dataset with our regression models to see if they could predict unknown fault locations. This showed us how well they handle new faults, and we found they need more work to be spot-on.



### 3.2 Design Method (PO(a))



**Step 1:** At first the Simulink VSC-HVDC transmission system model was built and modified

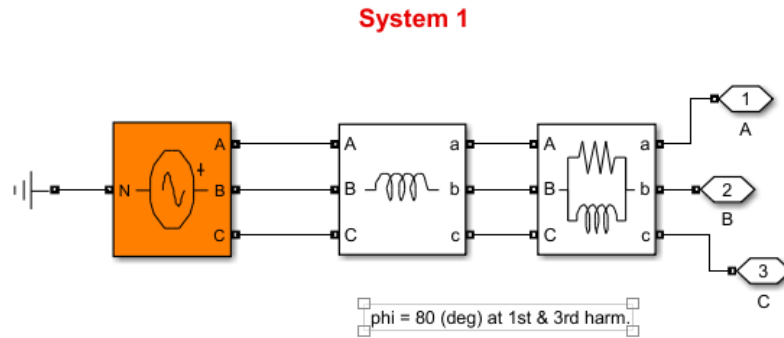
#### 1. Fundamental Theory & Power-Flow Equations

An HVDC link based on voltage-source converters (VSC-HVDC) enables independent control of active and reactive power between asynchronous AC systems. In steady state, the power transferred through the converter is governed by the classical equations:

$$P = -\left(\frac{U_s U_c}{X_c}\right) \sin \delta$$
$$Q = \frac{U_s (U_s - U_c \cos \delta)}{X_c}$$

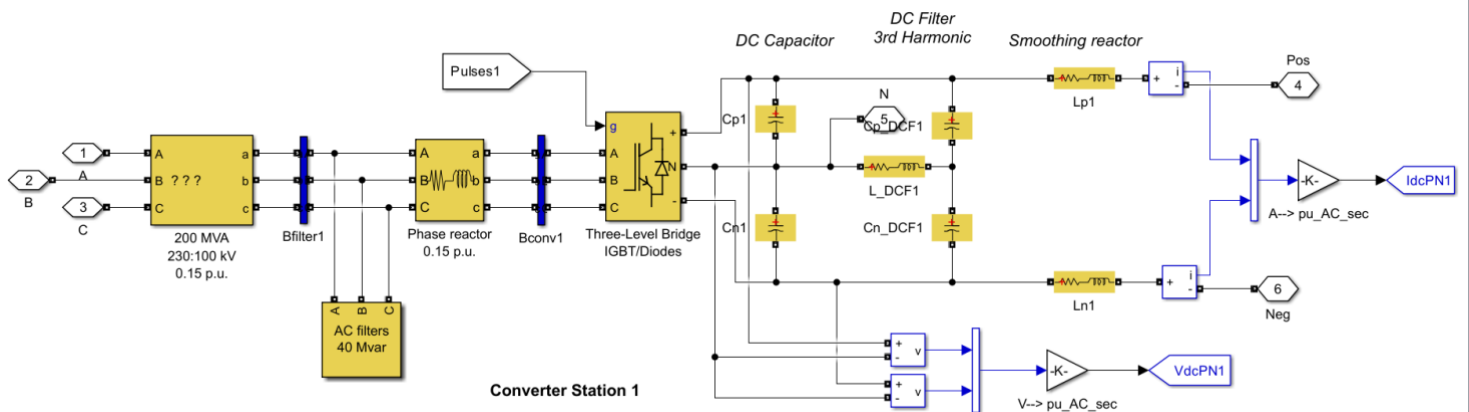
where  $U_s$  is the grid (AC source) voltage,  $U_c$  is the converter AC-side voltage,  $X_c$  is the series reactance (transformer + reactor), and  $\delta$  is the power-angle. These relations form the basis for both active-power control (via  $\delta$ ) and reactive-power control (via  $U_c$ ) in VSC converters.

## 2. AC System 1



**Figure:** AC System 1

An ideal three-phase voltage source (230 kV, 50 Hz) is used, with its neutral solidly grounded. Its outputs feed a per-unit-rated step-down transformer (230 kV  $\rightarrow$  100 kV) that matches the converter's DC-link voltage level. A 40 Mvar  $\pi$ -style AC filter is installed downstream to attenuate PWM harmonics, and a series reactor (0.15 pu) is provided to limit inrush and fault currents—thus ensuring grid stiffness and delivering clean waveforms for measurement.

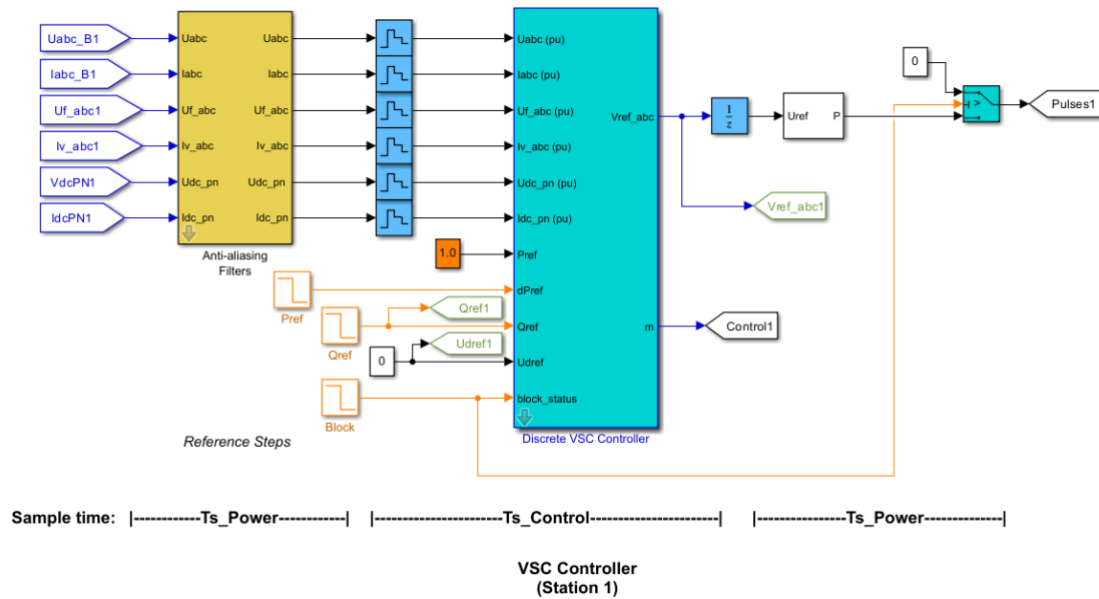


**Figure:** Station 1

## 3. Converter Station 1

The rectifier is modeled as a three-level IGBT/diode bridge whose DC terminals connect to a capacitor bank (Cp1/Cn1) holding 1.0 pu nominal voltage. A third-harmonic DC filter (L\_DCF1, C\_DCF1) is placed across the link to shunt high-order harmonics, and a series smoothing reactor (Lp1/Ln1) is inserted before the “Pos”/“Neg” buses to damp transient currents. Post-reactor voltage (VdcPN1) and current (IdcPN1) measurements are taken via gain blocks (pu  $\rightarrow$  V/A) and routed to the Data Acquisition subsystem.

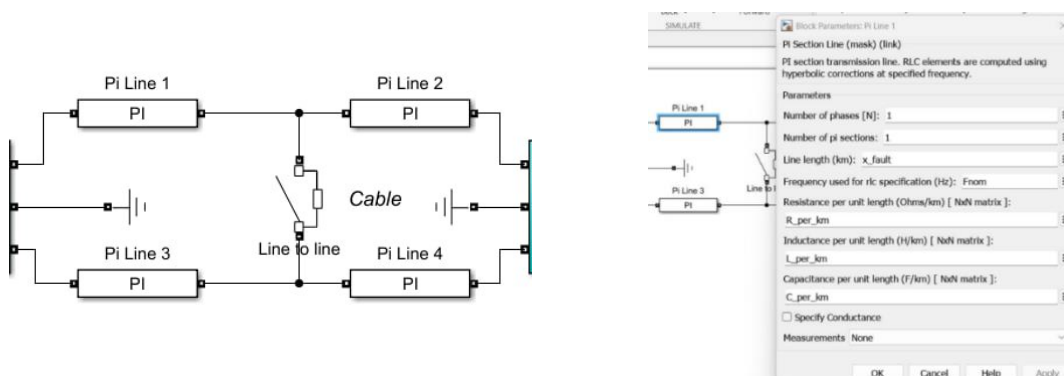
## 4. Control Subsystem



**Figure:** VSC Controller Station 1

- Phase-Locked Loop (PLL):** A three-phase PLL synchronizes the converter's modulation frame to the grid angle, ensuring accurate d/q transformation.
- Inner Current Regulator:** Fast PI controllers track  $i_{od}$  and  $i_{oq}$  references, decoupling active and reactive currents.
- Outer Loop (P/Q or Vdc Control):**
  - For the rectifier (Station 1), a DC-voltage regulator maintains  $V_{dc\_ref}$  by adjusting the d-axis current setpoint.
  - For the inverter (Station 2), an active/reactive power controller generates  $I_{d\_ref}$  and  $I_{q\_ref}$  from  $P_{ref}$  and  $Q_{ref}$  setpoints. A constant-voltage controller (Figure 3) is employed on the DC link to balance power and maintain steady DC voltage under varying load.

## 5. DC Link & $\pi$ -Section Modeling

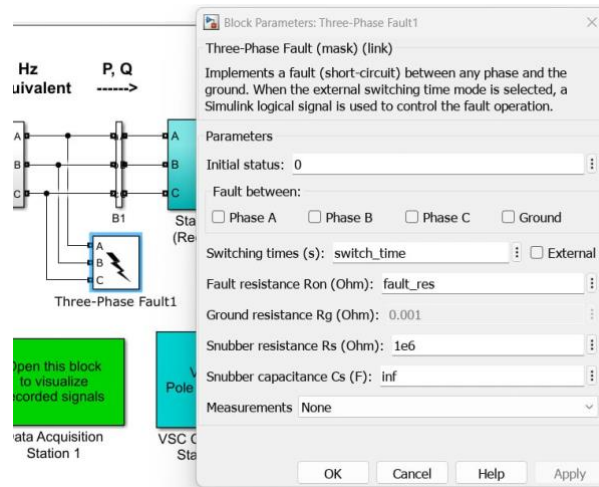


**Figure:** DC link and pi section modeling

Two cascaded  $\pi$ -section line blocks approximate the distributed R, L, and C of the 75 km DC

cable. (A central Breaker block simulates line faults by switching a low-resistance path at programmable times. This lumped-parameter approach balances physical fidelity with simulation speed across multiple fault scenarios.

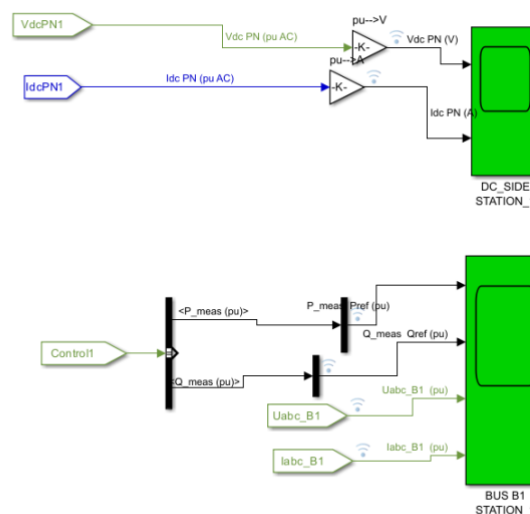
## 6. Fault Injection Blocks



**Figure:** Three phase Fault

- **AC Faults (Station 1 & Station 2):** Two Three-Phase Fault mask blocks allow programmable LG, LL, LLG, and LLL faults. Fault resistance (fault\_res) and switching time (switch\_time, switch\_time\_dis) are driven from the MATLAB workspace.
- **DC Faults:** The DC Breaker mask block simulates line-to-line shorts at user-selected positions (x\_fault) with breaker resistance and switch\_time\_dis and switch\_time parameters.

## 7. Measurement & Data Acquisition



**Figure:** Data acquisition Station 1

At each AC bus, three phase-voltages (Uabc) and three phase-currents (Iabc) are selected, along with converter P\_meas and Q\_meas signals. On the DC side, Vdc and Idc measurements are tapped post-reactor. All signals are funneled through anti-aliasing filters, then into Simulink's logsout via consolidated Data Acquisition blocks for post-processing.

### Step 2: Necessary environment –

Necessary signals which will be investigated were logged. So that after each simulation, the data of that signal will be saved in logged data.

Those are – 3 phase voltage, 3 phase current, real power, and reactive power on both AC system (total 16 signals), positive and negative voltage and current of DC transmission line on both bus side (total 8 signals). In total 24 signals were to be investigated.

As different fault conditions had different fault connection, fault resistance. All of these parameters were set with variable that would be controlled by the matlab code saved in the same workspace.

### Matlab code for parameters:

```
L_total = 75;
x_fault = 70;
R_per_km = 1.3900e-002;
L_per_km = 1.5900e-004;
C_per_km = 2.3100e-007;

switch_time = 1.5;
switch_time_dis = 10;
fault_res = 1000;

P_ref_var = 1.05;
Q_ref_var = -0.25;
```

Then a matlab code was built to extract data from the simulation and saving it for future work. The code structure is as followed:

### Matlab Code for Data Extraction:

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>% 1) Parameters tStart = 1.3; % windowing start time (s) dtWindow = 0.025; % window length (s) tEnd = 2.5; % windowing end time (s)  % 2) Grabbing logsout logs = out.logsout; nSigs = logs.numElements;  % 3) Time edges for windows edges = tStart:dtWindow:tEnd; % 1.3, 1.325, ..., 2.5 nWins = numel(edges)-1; % number of windows  % 4) Preparing feature names by probing first window firstSig = logs.get(1); Tfull = firstSig.Values.Time; Yfull = firstSig.Values.Data; nChans = size(Yfull,2);</pre> | <pre>% 5) Preallocating output % Columns: WindowIdx, WindowEndTime, then all RMS features data = zeros(nWins, 2 + numel(featsNames));  % 6) Computing RMS for each window for w = 1:nWins     t0 = edges(w);     t1 = edges(w+1);     % recording window end time     data(w,1) = w;     data(w,2) = t1;      col = 3; % column index in data     for i = 1:nSigs         sig = logs.get(i);         T = sig.Values.Time;         Y = squeeze(sig.Values.Data); % [nTime x nCh]</pre> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> featNames = {}; for i = 1:nSigs     sig = logs.get(i);     base = matlab.lang.makeValidName(sig.Name);     nC = size(sig.Values.Data,2);     for c = 1:nC         featNames{end+1} = sprintf('%s_C%d_RMS', base, c);     end end </pre> | <pre> mask = (T &gt;= t0) &amp; (T &lt; t1); for c = 1:size(Y,2)     ywin = Y(mask,c);     data(w,col) = sqrt(mean(ywin.^2)); % RMS     col = col + 1; end end  % 7) Building table and export varNames = [{'WindowIdx','WindowEnd_s'}, featNames]; T = array2table(data, 'VariableNames', varNames); writetable(T, 'DC fault_3.xlsx');  fprintf('Exported %d windows × %d features → HVDC_RMS_25ms_1p3_2p5.xlsx\n', ... nWins, size(data,2)); </pre> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Step 3:** The parameters code was run, then the Simulink model was simulated and after that the data extraction code was ran to extract data and save them.

**Step 4:** The extracted data was saved in a separate folder as mentioned.

**Step 5:** Then we setup the fault labels and the overall structure of the classification dataset by using binary numbers to denote the faults.

**Step 6:** The next step was to organize the final dataset by merging all the individual data according to the fault. Thus, the classification dataset was built.

**Step 7:** To build the fault location detection dataset, manual extraction of DC fault data were extracted from the classification dataset.

**Step 8:** A completely new dataset was built as test dataset for location detection problem.

**Step 9:** While the idea of using ANN, Random Forest, different CNN architecture were obtained from previous studies. The idea of 3 stage ANN and CNN classifier were built in order to improve the accuracy. All of the models are described below :

### 1. Single-Stage ANN

- A feed-forward neural network that takes the full 40-column RMS+label dataset, applies dense layers with dropout and batch normalization, and outputs a 14-way softmax for direct fault-type classification in one shot.

### 2. Random Forest Classifier

- An ensemble of 200 decision trees trained on the tabular RMS+Rf\_Ohm features, using bootstrap aggregation and feature sub-sampling to robustly partition the feature space and vote on the 14 fault classes.

### 3. Differing CNN Architectures

- Three variants of 1D-CNNs applied to the 48-window sequence of RMS features:
  - **Small Fixed Kernels:** Stacks of Conv1D(kernel=3) + pooling + dense layers
  - **Larger Kernels & Depth:** Conv1D(kernel=5) blocks with deeper fully connected heads
  - **Multi-Scale Kernels:** Parallel Conv1D branches (kernels=3,5,7) merged before dense layers to capture both short- and mid-range temporal patterns.

#### 4. Regression Models for Fault-Location

- Four regressors trained to predict fault position along the DC line:
  - **Polynomial Ridge (with interactions),**
  - **Stepwise Linear Regression,**
  - **MLPRegressor (dense NN),**
  - **Gaussian Process Regressor,**

Each is trained on the cleaned feature set and compared by MAE and  $R^2$  to select the best predictor.

#### 1. 3-Stage Hierarchical ANN Pipeline:

- **Stage 1 – Gate (Normal vs. AC vs. DC)**
  - **Input:** Full feature vector (40 dims).
  - **Feature Selection:** Swarm-optimized ANN uses GWO to pick the most discriminative subset.
  - **Architecture:** Dense(128→64) + BatchNorm + Dropout(0.3) → Softmax(3).
  - **Output:** 3-class probabilities (Normal/AC/DC).
- **Stage 2a – AC Fault Subtypes**
  - **Input:** RMS features + Rf\_Ohm for samples flagged “AC.”
  - **Feature Selection:** PSO selects best AC-relevant channels (U, I, P, Q).
  - **Architecture:** Dense(128→64) + BatchNorm + Dropout(0.3) → Softmax(8).
  - **Output:** One of 8 AC fault types.
- **Stage 2b – DC Fault Subtypes**
  - **Input:** RMS features + engineered DC features ( $\Delta I_{dc}$ ,  $\Delta V_{dc}$ , ratios) for “DC” samples.
  - **Feature Selection:** CSO chooses most informative DC channels and engineered metrics.
  - **Architecture:** Dense(128→64) + BatchNorm + Dropout(0.3) → Softmax(5).
  - **Output:** One of 5 DC fault locations.
- **Inference Flow**
  - A sample is first routed through Stage 1; based on that decision, it is passed to either Stage 2a or 2b for final classification.

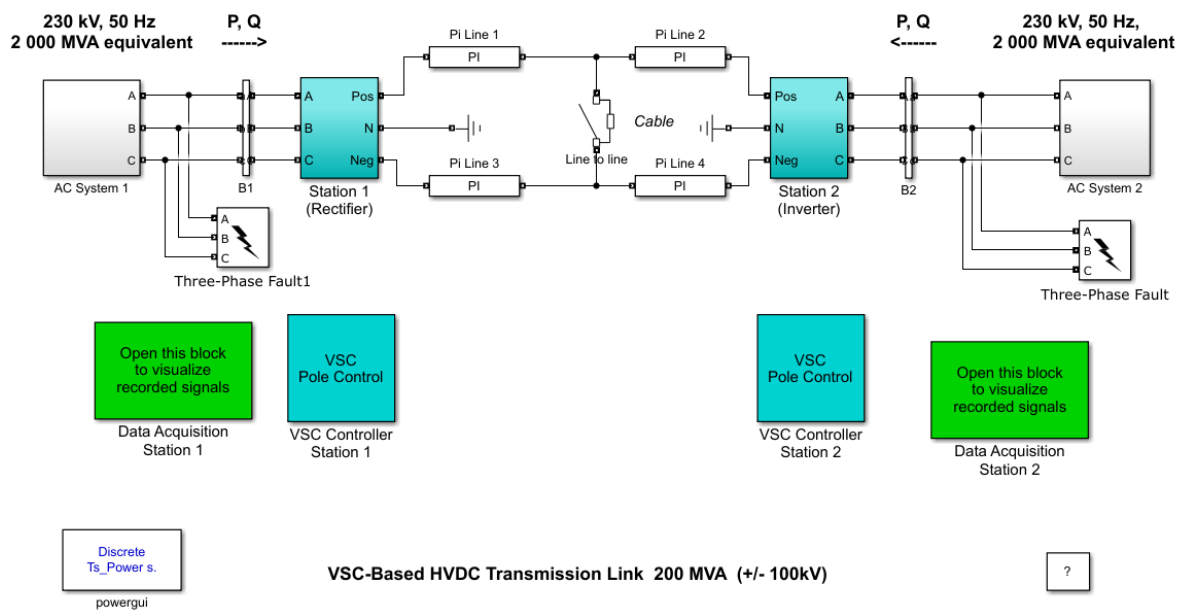
#### 2. 3-Stage Hierarchical CNN Pipeline:

- **Stage 1 – Gate CNN**
  - **Input:** 48×24 RMS feature matrix (time steps × channels).
  - **Architecture:** Conv1D(32, k=5) → MaxPool → Conv1D(64, k=5) → GlobalAvgPool → Dense(64) → Softmax(3).
  - **Output:** Normal/AC/DC probabilities.
- **Stage 2a – AC Fault CNN**
  - **Input:** Subset of the same 48×N\_AC matrix for AC-flagged windows.
  - **Architecture:** Two Conv1D layers (64→128 filters, k=3) + Dropout → LSTM(64) → Dense(64) → Softmax(8).
  - **Output:** One of 8 AC fault types.
- **Stage 2b – DC Fault CNN**
  - **Input:** 48×N\_DC matrix (only DC-relevant channels + engineered DC

features tiled along time).

- **Architecture:** Conv1D(32, k=3) → Conv1D(64, k=3) → Dropout → LSTM(32) → Dense(32) → Softmax(5).
- **Output:** One of 5 DC fault locations.
- **Inference & Metrics**
  - Each stage is trained separately with stratified splits and class weighting. During prediction, raw RMS windows flow sequentially from Stage 1 to the appropriate Stage 2 branch, and final labels are aggregated for overall accuracy and confusion-matrix analysis.

### 3.3 Simulation Model



**Figure:** VSC HVDC Simulink Model



## 4 Design Analysis and Evaluation

### 4.1 Novelty

#### 1. Dual Custom Datasets:

- Created **two separate datasets**:
  - A **classification dataset** with 14 classes (8 AC + 5 DC + no fault) and multiple fault resistances across 48 time windows.
  - A **regression dataset for fault location detection**, labeled with precise fault distances for DC faults, including variations in resistance and timing.
- This dual-dataset approach enables a **complete fault analysis (type + location)**, which is rarely addressed in a single framework.

#### 2. Three-Stage Fault Classification:

- Proposed a **hierarchical fault classification pipeline (No Fault → AC vs DC → Specific Fault Type)** to significantly improve **DC fault detection accuracy**.

### 4.2 Investigations (PO(d))

#### 4.2.1 Data Collection

##### • Classification Dataset:

With total 14 types of fault case (No fault, 8 AC faults, 5 DC faults), 10 varying fault resistance for each of the faults (except the No fault condition), using 10 P, Q combination for No fault, and using 48 window indices for each fault spanning from 1.3 to 2.5 sec. Fault occurring at 1.5 sec.

We got 6720 rows and for each row, we got 24 features from our simulation, adding up with fault resistance, window index and 14 fault types. We got total 40 columns.

Final dataset structure = 6720 \* 40.

| No Fault |          |      |       |       |      |      |       |       |      |       |       |       |       |       |        |           |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |       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|----------|----------|------|-------|-------|------|------|-------|-------|------|-------|-------|-------|-------|-------|--------|-----------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|------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| A        | B        | C    | D     | E     | F    | G    | H     | I     | J    | K     | L     | M     | N     | O     | P      | Q         | R       | S       | T       | U       | V       | W       | X       | Y       | Z       | AA      | AB      | AC      | AD      | AE      | AF      | AG      | AH      | AI      | AJ      | AK      | AL      |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |       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| 1        | No Fault | LG_R | LLG_R | LLL_R | LL_R | LG_S | LLG_S | LLL_S | LL_S | DC_L1 | DC_L2 | DC_L3 | DC_L4 | DC_L5 | Rf_Ohm | Windowwid | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | Uabc_B1 | U |

- **Location Detection Dataset:**

Now using distance of the fault in another column of the dataset, we built the dataset for DC location detection. For DC 5 fault location – 10 fault resistance – 48-time windows. In total 2400 rows.

Again 24 features with fault resistance, time window and distance(km).

Total 27 columns.

Final Dataset structure = 2400\*27.

|      | A      | B        | C         | D         | E         | F         | G         | H         | I         | J         | K         | L         | M         | N         | O         | P         | Q         | R         | S         | T         | U         | V         | W         | X         | Y         | Z         | AA        | AB        |
|------|--------|----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| 1    | Rt_Ohm | WindowId | Uabc_B1_g | Uabc_B1_g | Uabc_B1_g | Uabc_B1_g | Uabc_B1_g | Uabc_B1_g | Uabc_B1_g | Uabc_B1_g | Uabc_B1_g | Uabc_B1_g | Uabc_B1_g | Uabc_B1_g | Uabc_B1_g | Uabc_B1_g | Uabc_B1_g | Uabc_B1_g | Uabc_B1_g | Uabc_B1_g | Uabc_B1_g | Uabc_B1_g | Uabc_B1_g | Uabc_B1_g | Uabc_B1_g | Uabc_B1_g | Uabc_B1_g | Uabc_B1_g |
| 2    | 0.01   | 1        | 0.682538  | 0.73193   | 0.65751   | 0.708149  | 0.761201  | 0.683223  | 0.992443  | 0.002756  | 946.0042  | 948.3663  | 100528.5  | 100915.9  | 0.72942   | 0.756365  | 0.677182  | 0.625686  | 0.663357  | 0.594052  | 0.903137  | 0.100883  | 47274.08  | 47344.41  | 99463.52  | 99938.52  | 13        |           |
| 3    | 0.01   | 2        | 0.699932  | 0.6482    | 0.723379  | 0.727004  | 0.672665  | 0.752728  | 0.991749  | 0.003156  | 944.9933  | 948.1219  | 100757.1  | 100754.2  | 0.714641  | 0.686033  | 0.764572  | 0.62946   | 0.593667  | 0.661677  | 0.902319  | 0.098207  | 47190.09  | 47401.26  | 99718.52  | 99668.33  | 13        |           |
| 4    | 0.01   | 3        | 0.682233  | 0.731522  | 0.657452  | 0.709822  | 0.762215  | 0.685988  | 0.993565  | 0.002261  | 945.0376  | 951.8259  | 100826.2  | 100691.9  | 0.729559  | 0.756355  | 0.677558  | 0.626943  | 0.664805  | 0.596371  | 0.903534  | 0.098957  | 47346.33  | 47504.5   | 99752.41  | 99786.69  | 13        |           |
| 5    | 0.01   | 4        | 0.699901  | 0.648048  | 0.723469  | 0.728979  | 0.67412   | 0.731847  | 0.994776  | 0.001663  | 945.5026  | 950.5574  | 100791    | 100983.2  | 0.714767  | 0.686465  | 0.764538  | 0.632169  | 0.593966  | 0.663225  | 0.904291  | 0.100135  | 47157.76  | 47580.36  | 99842.64  | 99845.47  | 13        |           |
| 6    | 0.01   | 5        | 0.682385  | 0.731716  | 0.65735   | 0.711665  | 0.763798  | 0.686965  | 0.995084  | 0.001614  | 943.6157  | 949.7825  | 101033.3  | 100942.2  | 0.729594  | 0.756323  | 0.677078  | 0.629516  | 0.665414  | 0.596353  | 0.905277  | 0.099661  | 47211.81  | 47431.6   | 99961.98  | 99934.3   | 13        |           |
| 7    | 0.01   | 6        | 0.699604  | 0.647802  | 0.722939  | 0.73067   | 0.678093  | 0.755696  | 0.996659  | 0.00096   | 948.9099  | 950.2601  | 101152.5  | 100800.7  | 0.71458   | 0.68606   | 0.764432  | 0.633888  | 0.596224  | 0.667007  | 0.906909  | 0.097571  | 47422.28  | 47514.55  | 100136.5  | 99764.9   | 13        |           |
| 8    | 0.01   | 7        | 0.682226  | 0.731569  | 0.657197  | 0.712789  | 0.7651    | 0.687763  | 0.998922  | 0.000841  | 948.8275  | 950.1143  | 101114.2  | 100775.5  | 0.729635  | 0.756419  | 0.677301  | 0.631382  | 0.667992  | 0.597085  | 0.904666  | 0.097337  | 47437.36  | 47479.51  | 100243.1  | 99783.24  | 13        |           |
| 9    | 0.01   | 8        | 0.699485  | 0.64756   | 0.722891  | 0.733449  | 0.679417  | 0.756157  | 0.998057  | 0.003358  | 948.3349  | 950.9085  | 101115.7  | 100982.2  | 0.714574  | 0.686322  | 0.764499  | 0.634151  | 0.597174  | 0.665507  | 0.908237  | 0.098467  | 47354.81  | 47556.67  | 100136.2  | 99679.86  | 13        |           |
| 10   | 0.01   | 9        | 0.458963  | 0.503086  | 0.45691   | 2.637015  | 2.183234  | 1.930708  | 0.834663  | 0.927287  | 7564.757  | 8610.418  | 17638.79  | 17373.47  | 0.544631  | 0.593262  | 0.523092  | 2.044109  | 1.632749  | 2.234223  | 0.523708  | 0.841309  | 294425.6  | 234612    | 22953.41  | 20801.9   | 13        |           |
| 2391 | 2000   | 38       | 0.708112  | 0.656491  | 0.732707  | 0.64629   | 0.6127    | 0.682844  | 0.902777  | 0.096904  | 852.2622  | 858.0089  | 101308.3  | 100710    | 0.715325  | 0.683933  | 0.762535  | 0.507196  | 0.471638  | 0.528403  | 0.71728   | 0.102692  | 37625.69  | 37874.82  | 100403    | 99779.98  | 65        |           |
| 2392 | 2000   | 39       | 0.691972  | 0.740825  | 0.665435  | 0.651069  | 0.683322  | 0.61085   | 0.901914  | 0.097455  | 856.3209  | 855.8629  | 100930.6  | 100916.5  | 0.726463  | 0.755855  | 0.676687  | 0.497746  | 0.530587  | 0.475941  | 0.718149  | 0.098309  | 37816.48  | 37761.69  | 99988.96  | 100043.4  | 65        |           |
| 2393 | 2000   | 40       | 0.707963  | 0.656405  | 0.73295   | 0.645581  | 0.610624  | 0.682036  | 0.901178  | 0.100092  | 852.9626  | 857.3557  | 100507.5  | 101161.4  | 0.715347  | 0.68406   | 0.762481  | 0.506686  | 0.471202  | 0.526896  | 0.716937  | 0.09752   | 37651.05  | 37859.24  | 99631.68  | 100185.5  | 65        |           |
| 2394 | 2000   | 41       | 0.691961  | 0.741162  | 0.665347  | 0.650331  | 0.683819  | 0.613048  | 0.901364  | 0.099924  | 856.4165  | 852.8435  | 100536.2  | 101354.2  | 0.726863  | 0.756418  | 0.677144  | 0.496658  | 0.531006  | 0.475865  | 0.715678  | 0.102581  | 37755.86  | 37625.38  | 99613.85  | 100473    | 65        |           |
| 2395 | 2000   | 42       | 0.708135  | 0.656695  | 0.732901  | 0.645631  | 0.611828  | 0.682562  | 0.901925  | 0.101089  | 856.1095  | 855.0421  | 100369.6  | 101384.1  | 0.715271  | 0.684066  | 0.762569  | 0.50717   | 0.472023  | 0.527281  | 0.717251  | 0.101979  | 37804.77  | 37739.08  | 99425.55  | 100455.4  | 65        |           |
| 2396 | 2000   | 43       | 0.691939  | 0.740894  | 0.665583  | 0.652176  | 0.682285  | 0.610762  | 0.901763  | 0.099552  | 854.4637  | 854.8633  | 100692.9  | 101101.7  | 0.726648  | 0.755935  | 0.676804  | 0.497566  | 0.530375  | 0.477004  | 0.717424  | 0.099353  | 37672.31  | 37743.05  | 99803.26  | 100240.5  | 65        |           |
| 2397 | 2000   | 44       | 0.70826   | 0.656642  | 0.733011  | 0.644053  | 0.610412  | 0.680721  | 0.900157  | 0.100859  | 851.741   | 854.6022  | 101182.2  | 100575.9  | 0.715001  | 0.683723  | 0.762377  | 0.503577  | 0.46967   | 0.534739  | 0.715727  | 0.098078  | 37554.65  | 37761.02  | 100277.3  | 99618.95  | 65        |           |
| 2398 | 2000   | 45       | 0.692007  | 0.741035  | 0.665609  | 0.649086  | 0.681242  | 0.609902  | 0.899388  | 0.101053  | 850.9833  | 854.4327  | 101468.3  | 100321.8  | 0.726834  | 0.756267  | 0.677011  | 0.497619  | 0.529791  | 0.475648  | 0.71411   | 0.099055  | 37542.17  | 37691.35  | 100566.1  | 99431.9   | 65        |           |
| 2399 | 2000   | 46       | 0.70806   | 0.656536  | 0.73287   | 0.646104  | 0.61276   | 0.682726  | 0.900145  | 0.098968  | 854.6293  | 857.8741  | 101261.9  | 100528.3  | 0.715222  | 0.684045  | 0.762573  | 0.505379  | 0.470686  | 0.525791  | 0.715459  | 0.101128  | 37763.62  | 37844.74  | 100372.3  | 99620     | 65        |           |
| 2400 | 2000   | 47       | 0.692164  | 0.741168  | 0.665773  | 0.648964  | 0.680327  | 0.610268  | 0.901451  | 0.101444  | 856.0037  | 853.6366  | 100948    | 100938.2  | 0.726663  | 0.756179  | 0.676955  | 0.497376  | 0.53154   | 0.476118  | 0.716875  | 0.100614  | 37765.56  | 37697.77  | 99904.74  | 100012.9  | 65        |           |
| 2401 | 2000   | 48       | 0.707965  | 0.656511  | 0.732996  | 0.644975  | 0.609969  | 0.681021  | 0.899     | 0.100915  | 852.0091  | 850.1794  | 100591.1  | 101185.8  | 0.715087  | 0.683831  | 0.762351  | 0.504519  | 0.468673  | 0.523753  | 0.71453   | 0.099147  | 37622.19  | 37472.43  | 99661.77  | 100276.3  | 65        |           |

## 4.2.2 Results:

### Classification

#### 1. ANN:

| Model                                    | Test Accuracy |
|------------------------------------------|---------------|
| Dense-only, batch 128, 100 epochs        | 64.3%         |
| Dense only, batch 128, 500 epochs        | 67.85%        |
| Dense-only, batch 512, 500 epochs        | 69.74%        |
| Added Dropout (0.3) and extra Dense (64) | 68.75%        |

#### 2. Random Forest:

Test Accuracy: 73.74%

#### 3. CNN:

| Model                                               | Test accuracy |
|-----------------------------------------------------|---------------|
| Simple CNN with uniform small kernels (3)           | 62.57         |
| CNN with larger kernels (5) and bigger dense layers | 62.28         |
| Multi-scale CNN with varied kernel sizes (3, 5, 7)  | 66            |

#### 4. 3 stage hierarchical stage:

##### ANN:

| Model and algorithm      | Test accuracy |
|--------------------------|---------------|
| Normal vs AC vs DC - GWO | 92.6          |
| AC classification - PSO  | 75.13         |
| DC classification - CSO  | 54.37         |

##### CNN:

| Stage              | Test accuracy |
|--------------------|---------------|
| Normal vs AC vs DC | 93.15         |
| AC classification  | 86.57         |
| DC classification  | 80.18         |

### Location Detection

| Length<br>(Km) | Interactive<br>Linear<br>Regression<br>(Predicted – accuracy) | Stepwise linear<br>regression<br>(Predicted – accuracy) | Neural<br>Network<br>(Predicted – accuracy) | Gaussian<br>Process<br>regression<br>(Predicted – accuracy) |
|----------------|---------------------------------------------------------------|---------------------------------------------------------|---------------------------------------------|-------------------------------------------------------------|
| 10             | 36 – 64%                                                      | 14 – 72%                                                | 33 – 68%                                    | 22 – 78%                                                    |
| 24             | 39 – 78%                                                      | 19 – 58%                                                | 37 – 80%                                    | 30 – 89%                                                    |
| 35             | 40 – 91%                                                      | 20 – 51%                                                | 40 – 90%                                    | 35.67 – 94%                                                 |
| 48             | 41 – 87%                                                      | 20 – 43%                                                | 44 – 89%                                    | 41 – 87%                                                    |
| 60             | 40 – 73%                                                      | 20 – 36%                                                | 48 – 81%                                    | 47 – 80%                                                    |
| 70             | 42 – 63%                                                      | 20 – 30%                                                | 50 – 73%                                    | 52 – 74%                                                    |

## 4.2.3 Interpretation and Conclusions on Data

### Interpretation:

#### 1. Flat ANN Performance

- The baseline dense network (128 units) converged to only ~64 % accuracy after 100 epochs, improving to ~69.7 % when trained longer (500 epochs) and with added dropout/dense layers.
- Flattening 48-window time series into a single vector discards temporal ordering. The network can still learn feature correlations but lacks inductive bias for sequence dynamics (e.g. the way a fault signature evolves over successive windows).

#### 2. Random Forest Strengths

- Achieved ~73.7 % test accuracy on the 13-class task—surpassing all single-stage neural nets.
- Tree ensembles naturally handle heterogeneous tabular inputs (RMS magnitudes, fault resistance, window index) and robustly partition the feature space without requiring extensive hyperparameter tuning or feature scaling.

#### 3. 1D-CNN Variants

- **Uniform small kernels (size=3):** ~62.6 % accuracy.
- **Larger kernels & deeper dense layers:** ~62.3 %—no improvement, indicating overparameterization.
- **Multi-scale kernels (3,5,7):** ~66 %, the best among CNNs but still below Random Forest.
- While multi-scale convolutions capture local temporal patterns (short- vs. mid-term RMS transitions), the downsampled RMS features provide limited sequential granularity (only 48 points), capping CNN gains.

## 4. Three-Stage Hierarchical Architectures

- **ANN-Based Hierarchy:**
  - **Stage 1 (Normal vs AC vs DC):** 92.6 % with GWO—excellent separation of healthy vs. fault group.
  - **Stage 2 (AC subtypes):** 75.1 % with PSO—moderate difficulty across LL, LG, LLG, LLL on two stations.
  - **Stage 3 (DC subtypes):** 54.4 % with CSO—poor performance, indicating that DC fault signatures in the RMS domain are poorly distinguished by a simple dense classifier.
- **CNN-Based Hierarchy:**
  - **Stage 1:** 93.2 %—slightly better than ANN, showing CNN’s ability to exploit even coarse temporal structure at the top level.
  - **Stage 2:** 86.6 %—a dramatic jump versus ANN’s 75 %, highlighting CNN’s advantage in classifying AC fault dynamics.
  - **Stage 3:** 80.2 %—a significant improvement over ANN’s 54 % and single-stage CNN (~66 %), demonstrating that when focused on DC-fault windows alone, temporal convolution helps isolate subtle DC-fault waveforms.

## Conclusion

- We simulated a VSC HVDC Transmission System using a Simulink-based model and generated a structured dataset for fault classification and fault location detection.
- Multiple ML and DL models were evaluated, with the **three-stage CNN classifier** emerging as the most effective, outperforming flat models and ANN hierarchies. This hierarchical CNN pipeline now forms the baseline for our fault classification approach.
- **DC-fault classification (Stage 3)** remains the weakest link. Enhancements such as:
  - Feature augmentation (time-derivatives like  $\Delta I_{dc}/\Delta t$ ,  $\Delta V_{dc}/\Delta t$ , or moving-window skewness/kurtosis),
  - Hybrid CNN-ML models (e.g., CNN→MLP→Random Forest),
  - Data balancing and augmentation (synthetic DC fault samples), can boost DC fault accuracy beyond 85%.
- For practical deployment, Stage 1 and Stage 2 can run on a single CNN backbone with lightweight stage-specific heads, while Stage 3 may require a separate low-latency microservice or FPGA-accelerated CNN block.
- For fault location detection, **Gaussian Process Regression** achieved better results compared to other models, though improvements are still required to achieve higher accuracy.
- In the future, one may aim to refine the dataset, test more complex hybrid models, and identify the reasons behind low accuracies to achieve **95%+ classification accuracy**.

For more accurate fault location prediction, we plan to enrich the dataset further and combine CNN-based feature extraction with advanced regression models, ensuring robust real-world HVDC protection performance.

### 4.3 Limitations of Tools (PO(e))

While machine learning (ML) and deep learning (DL) offer excellent ways of fault classification and detection, they are not without their drawbacks. The performance of an ML or DL model greatly depends upon the training dataset. The more diverse or robust the dataset is, the better the model performs in a practical case. In our project, we simulated the reference VSC HVDC model and took values for different parameters. As it required manual parameter handling, we had to record data for every parameter combination separately and in the end, combine them all together. This process was lengthy and repetitive, which prevented us from generating a vast dataset. As the dataset was limited, so was the model.

Moreover, these simulated data may not capture the actual values of the power system entirely. Furthermore, high computational requirements for training and deploying complex DL algorithms may not be feasible for real time applications in power systems with limited hardware resources. Finally, these tools require continuous maintenance, including periodic retraining to adapt to changing system conditions or new types of faults, which can increase operational complexity and costs. Any change made in the system will render the model ineffective, requiring the model to be trained and tuned again.

### 4.4 Impact Assessment (PO(f))

The implementation of machine learning (ML) and deep learning (DL) in fault classification and location detection for VSC-HVDC transmission systems presents a range of societal, cultural, health and safety, and legal implications.

#### 4.4.1 Assessment of Societal and Cultural Issues

The adoption of advanced automation and intelligent fault detection systems may influence the workforce in power systems by reducing reliance on manual monitoring and traditional engineering roles, which could lead to job displacement or necessitate reskilling of employees. On the positive side, improved reliability and rapid fault response can support societal progress by enhancing the stability of the electricity supply. An unhampered power supply is crucial for modern economies and daily life.

#### 4.4.2 Assessment of Health and Safety Issues

Utilizing ML/DL based systems for fault detection directly contributes to enhanced health and safety within power grid operations by enabling quicker response times to dangerous faults, potentially preventing equipment damage, fires, or large-scale outages. Automated detection systems can minimize human exposure to high risk environments by reducing the need for on site inspections after a fault occurs. Nevertheless, reliance on these tools introduces risks if they malfunction or misclassify faults.

#### **4.4.3 Assessment of Legal Issues**

The use of ML/DL algorithms introduces unique legal considerations, such as accountability in the event of system failures or incorrect fault diagnoses. Clearly defining liability among system operators, technology providers, and equipment manufacturers is essential, especially since software driven decisions may lack transparency. Legal frameworks may also need to adapt to address issues surrounding the intellectual property of ML models and cybersecurity for critical grids.

#### **4.5 Sustainability Evaluation (PO(g))**

Applying ML/DL driven fault classification and detection techniques improves the overall performance of the power distribution system. When faults occur, they are identified rapidly, resulting in a quicker response, decreasing maintenance time and blackout time. Efficient fault management also increases the lifespan of the equipment used in the power grid.

As our model works on the current, voltage, active power and reactive power of the system, it can be easily applied to the system from a control station. Occasional retraining and tuning of the model will make the system economically efficient and infrastructurally sustainable.

#### **4.6 Ethical Issues (PO(h))**

Ethical issues in applying machine learning and deep learning to fault classification and location detection in transmission systems primarily revolve around concerns of transparency, accountability, and data privacy. Since ML/DL algorithms often operate as "black boxes," their lack of transparency complicates understanding and trusting their decisions, raising accountability challenges if faults are misclassified or missed. Privacy issues also arise due to the extensive data collection needed for training. A strong safeguard is required against unauthorized access and misuse of the data. Moreover, reliance on automated systems demands maintaining human oversight to ensure ethical alignment and prevent harm from errors. Addressing these ethical challenges requires careful design, continuous monitoring, and responsible use of ML/DL technologies in power systems

## 5 Reflection on Individual and Team work (PO(i))

### 5.1 Individual Contribution of Each Member

| Different Segment of Project       | Individual Contribution                              |
|------------------------------------|------------------------------------------------------|
| Group Discussion                   | 2106176, 2106179, 2106178, 2106179, 2106180 ,2106181 |
| Creating VSC-HVDC Simulink model   | 2106176, 2106179, 2106181                            |
| Data Extraction & Dataset Creation | 2106177, 2106178                                     |
| Model Creation                     | 2106181, 2106176                                     |
| Feeding the dataset to the model   | 2106179, 2106180                                     |
| Training the model                 | 2106176, 2106179, 2106181                            |
| Evaluation                         | 2106177, 2106178, 2106180                            |
| Preparing Final Report             | 2106176, 2106179, 2106178, 2106179, 2106180 ,2106181 |



## 5.2 Mode of TeamWork

Our team consisted of six members. Each team member brought a unique set of skills, perspectives, and experiences.

**Project Planning:** Our team collectively decided on this project after thorough discussions regarding its scope, the anticipated data requirements, potential machine learning and deep learning approaches, and the overall architectural design. We then meticulously planned each software development module and Identify potential challenges related to data volume and quality, model complexity, computational resources and achieving target performance accuracies.

**Task Distribution:** Following a successful planning phase, we strategically distributed the diverse project tasks among the team members. This included assigning responsibilities for data collection, model development, algorithm selection and system integration.

**Software Implementation:** This project primarily involved software implementation. We developed and simulated various fault scenarios for HVDC transmission lines using a Simulink model to generate comprehensive and realistic datasets. This involved designing the electrical circuits and injecting various AC faults (LL, LG, LLG, LLL) and DC faults (LL, LG) at different locations and under varying conditions.

**Model Development and Training:** With the simulated data, we proceeded to develop and train Machine Learning and Deep Learning models. This involved data preprocessing, feature engineering, model selection (CNN, ANN, Random Forest for classification; Linear Regression, Neural Network, Gaussian Process Regression for regression), and model training and optimization.

**Testing and Validation:** We rigorously tested and validated our developed software models to ensure their accuracy and reliability. This involved evaluating the models using performance metrics and testing against unseen fault scenarios.

**Communication and Discussion:** Throughout the project lifecycle, we maintained open and continuous communication. We held multiple group discussions, fostering an environment where every team member felt empowered to voice their opinions. All major decisions were made collaboratively, following thorough discussions and consensus among the team members.

## 5.3 Diversity Statement of Team

Every one of the team members have their own skill and expertise. We combined all the skills and expertise to complete our project. Whenever a problem arose, we were able to solve it because of the diverse knowledge of the team.

## 5.4 Log Book of Project Implementation

| Date       | Milestone achieved                                                                          | Individual Role                                                                                                                         | Team Role                                                                                        | Comments                                 |
|------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------------|
| 29/04/2025 | Group Discussion                                                                            | 2106176, 2106179, 2106178, 2106179, 2106180, 2106181<br>Joined the group discussion                                                     | As a team we focused on the theoretical background of the project and completed project planning |                                          |
| 12/05/2025 | Creating VSC-HVDC Simulink model                                                            | 2106176, 2106179, 2106181 Created VSC-HVDC Simulink model using MATLAB                                                                  | Other team member gave their advice                                                              |                                          |
| 09/06/2025 | Data Extraction & Dataset Creation for classification                                       | 2106177, 2106178 extracted data and created dataset                                                                                     | All members were present so that we could solve any problem we face easily                       |                                          |
| 20/06/2025 | Model Creation and training the model and testing the accuracy                              | 2106181, 2106176 Coded the architectures of chosen ML/DL models (ANN, Random Forest, CNN for classification, trained and test the model | Other team member gave their advice                                                              |                                          |
| 04/07/2025 | Dataset Creation for detecting fault location                                               | 2106179, 2106180 extracted data and created dataset for detecting location                                                              |                                                                                                  |                                          |
| 15/07/2025 | Model Creation for fault location detection and training the model and testing the accuracy | 2106176, 2106179, 2106181 build model and trained, test the models using the labeled data                                               |                                                                                                  |                                          |
| 18/07/2025 | Evaluation                                                                                  | 2106177, 2106178, 2106180 evaluate model performance using appropriate metrics                                                          | All members were present so that we could solve any problem we face easily                       | We tried our best to find the best model |
| 24/07/2025 | Preparing Final Report                                                                      | 2106179 divided the report writing                                                                                                      | Other member completed their                                                                     |                                          |

|  |  |                                                                                            |                                                          |  |
|--|--|--------------------------------------------------------------------------------------------|----------------------------------------------------------|--|
|  |  | tasks for all member and compiled it after collecting different part from different member | respected portion of report within the given time period |  |
|--|--|--------------------------------------------------------------------------------------------|----------------------------------------------------------|--|

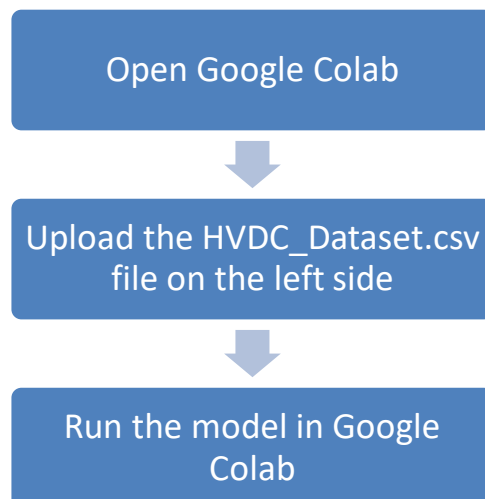
## 6 Communication to External Stakeholders (PO(j))

### 6.1 Executive Summary

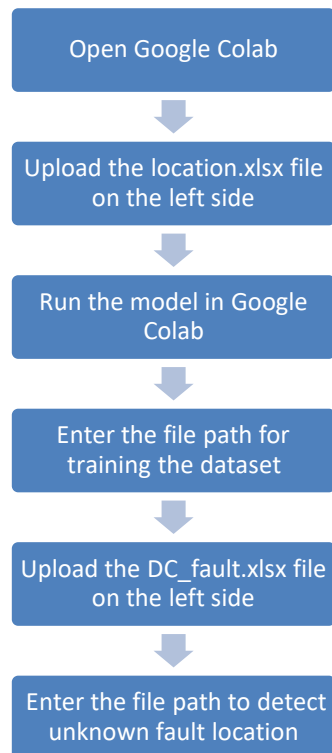
The project aims to develop a reliable fault classification and localization method for HVDC systems using machine learning. A simulated model of a mimicking VSC HVDC system was created in Simulink to generate fault data. Multiple machine learning and deep learning models were evaluated, with a three-stage Convolutional Neural Network (CNN) classifier achieving the best performance, with classification accuracies exceeding 80%. While the solution shows promising progress in detecting and classifying faults, especially in AC components, classification of DC faults remains challenging. This work marks a significant step towards enhancing HVDC system fault detection with machine learning, with potential to improve grid reliability and fault management.

### 6.2 User Manual

#### Classification:



## Regression:



## 7 Project Management (PO(k))

Our project, "Fault Classification and Location Detection of a VSC-HVDC Transmission System using ML and DL," inherently involved robust project management principles to ensure its successful completion, aligning directly with **Program Outcome PO(k): "Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments."** This was demonstrated through meticulous planning, strategic task distribution, and continuous oversight of resources and progress.

Key aspects of project management and implied financial considerations included:

- **Structured Project Planning:** From the outset, we collectively defined the project's scope, objectives, and the technical requirements, including the identification of specific features for our DNN models. This initial planning phase laid the groundwork for efficient resource allocation and minimized redundant efforts, embodying sound engineering management principles.
- **Strategic Task Distribution:** Following comprehensive planning, tasks were strategically distributed among the six team members. This ensured that specialized responsibilities, such as Simulink model modification, data extraction, dataset creation for both classification and regression, and the implementation of various ML/DL models (e.g., ANN, CNN, Random Forest, various regression models), were assigned based on individual strengths and to balance workload. This efficient task allocation optimized the use of human resources, a key aspect of project management.

- **Time Management & Workflow Adherence:** The project adhered to a structured timeline, evident in the sequential workflow. Early phases were dedicated to feature definition and Simulink model adaptation. Subsequent periods focused intensely on data extraction and the meticulous creation of comprehensive datasets for both classification and regression. Core machine learning activities including model creation, feeding, training, and initial evaluation were then systematically carried out. This clear phasing helped in managing time as a critical project resource.
- **Resource Optimization:** Although direct financial costs were not tracked in detail, the project involved effective management of computational resources (Simulink, MATLAB, ML/DL frameworks) and team expertise. By defining features early and modifying the Simulink model precisely, we optimized the data generation process, which could otherwise be computationally intensive. The choice and experimentation with various ML/DL models also demonstrated an understanding of leveraging available tools to achieve desired outcomes efficiently.
- **Continuous Evaluation and Adaptation:** The iterative process of initial model evaluation, followed by advanced optimization and in-depth analysis, showcased adaptive project management. When initial models didn't meet accuracy targets (e.g., Random Forest at 73.74%), the team developed and implemented more complex solutions like the 3-stage hierarchical CNN, demonstrating flexibility in approach and a commitment to achieving project goals within resource constraints.
- **Risk Management & Problem Solving:** Identifying limitations, such as low accuracy in DC fault classification or inconsistent regression performance, and subsequently proposing clear future work (rechecking datasets, using more complex/hybrid models) reflected an understanding of project risks and a proactive approach to continuous improvement.

In essence, the project's execution, from defining technical requirements and managing workflow to optimizing model performance, collectively demonstrated effective engineering management principles and economic decision-making, fulfilling the requirements of PO(k) within a complex, multidisciplinary problem-solving environment.

## 8 Future Work (PO(l))

- **Improve DC Fault Classification:** Enhance the detection accuracy of DC faults using hybrid deep learning models (e.g., CNN-LSTM or attention-based models) to better capture weak DC fault signatures.
- **Advance Regression for Fault Location:** Develop more accurate fault location models by combining CNN with Gaussian Process Regression or LSTM to reduce location errors, especially under high-resistance DC faults.
- **Dataset Expansion:** Include more diverse fault conditions (different locations, resistances, and converter settings) and introduce noise variations for better real-world performance.
- **Real-Time Implementation:** Test the models on real-time platforms (e.g., OPAL-RT or hardware-in-the-loop) to evaluate speed and reliability.
- **Multi-Terminal HVDC Extension:** Adapt the approach for fault detection and location in multi-terminal VSC-HVDC networks.

## 9 References

1. H. N. Bhadauria and R. Kumar, "Protection of an HVDC transmission system using wavelet transform and S transform," *Protection and Control of Modern Power Systems*, vol. 5, no. 1, p. 20, 2020.
2. M. H. Bollen and F. Hassan, "Advanced Control Strategies of VSC Based HVDC Transmission System: Issues and Potential Recommendations," ResearchGate, 2018.
3. K. R. Jha and P. K. Jha, "Detection, classification and location of transmission line faults using Discrete Wavelet Transform," ResearchGate, 2021.
4. S. K. Singh, R. K. Singh, and R. P. Saini, "Fault Detection and Classification in HVDC Transmission System: A Review," *Sensors*, vol. 20, no. 16, p. 4438, 2020.
5. S. Lan, D. Chen, and H. Xu, "A novel HVDC double-terminal non-synchronous fault location method based on 1D-CNN," *IEEE Transactions on Power Delivery*, vol. 34, no. 3, pp. 1051–1062, Jun. 2019. doi: 10.1109/TPWRD.2019.2892135.
6. C. Ding, Z. Wang, and Z. Yuan, "Convolutional neural network based on fast Fourier transform and Gramian angular field for fault identification of HVDC transmission line," *Sustainable Energy, Grids and Networks*, vol. 32, p. 100902, Dec. 2022. doi: 10.1016/j.segan.2022.100902.
7. M. A. Hossain, T. Islam, and R. K. Paul, "Internal and external DC fault discrimination in VSC-HVDC system using Stockwell transform and bi-directional GRU," *Electric Power Systems Research*, vol. 227, p. 109786, 2024. doi: 10.1016/j.epsr.2024.109786.
8. R. S. Jawad and H. Abid, "Fault detection and classification for VSC-HVDC systems by using different swarm optimization algorithms-based neural network," *Engineered Science*, vol. 23, pp. 213–226, 2023. doi: 10.30919/es8e1008.
9. M. A. A. Khan et al., "Fault Diagnosis Using Deep Learning Methods in Electrical Power Systems," *IEEE Access*, vol. 8, pp. 157562–157579, 2020.
10. S. Agarwal et al., "Classification of HVDC Faults Using Convolutional Neural Networks," ES Publisher, 2020.
11. J. Wang et al., "A Multi-Stage Deep Learning-Based Classifier for HVDC Fault Detection," *IEEE Transactions on Power Delivery*, 2020.
12. H. Li, Z. Zhang, and B. Wu, "Recent Advances in Fault Location Techniques for HVDC Transmission Systems," *Processes*, vol. 13, no. 2, p. 527, 2025.